

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO1

:Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 20%

Total Marks:50

Annexure A

QUESTIONS: 5 *Code of Conduct:*

CONCEPT MAPPING

I B.Munna/192372028 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

B.Munna

Annexure A

CONCEPT MAPPING

Security Attacks, Threats, Risks and

1. Create a concept map linking the following:

Security Goals (Confidentiality, Integrity, Availability. Perform the following tasks:

a. Show how each attack leads to specific threats and risks

b. Map how security goals help mitigate them Provide one real-world example for each link

2. Develop a concept map comparing substitution techniques: Caesar Cipher, Monoalphabetic Cipher, Vigenère Cipher and Hill Cipher. Perform the following tasks to connect based on: Key usage, Complexity and Security strength. Analyze which technique is most secure and why.

3. Construct a concept map for Classical Encryption Schemes: Substitution Techniques and Transposition Techniques. Perform the following tasks: a. Show differences and similarities

i. Map how each affects the Data structure and Security level

ii. Analyze which is more vulnerable to attacks and why

4. Create a concept map connecting Steganography concepts: Data Hiding, Watermarking and Survivability. Perform the following tasks:

a. Show relationships between these concepts

b. Analyze how each contributes to secure communication

c. Include one real-world application for each

5. Draw a concept map linking number theory concepts used in cryptography: Modular Inverse, Extended Euclid Algorithm, Fermat's Little Theorem, Euler's Phi Function and Euler's Theorem. Perform the following tasks:

a. Show how each concept is mathematically connected

b. Map their role in encryption/decryption

c. Analyze which concept is most critical for public key cryptography and justify

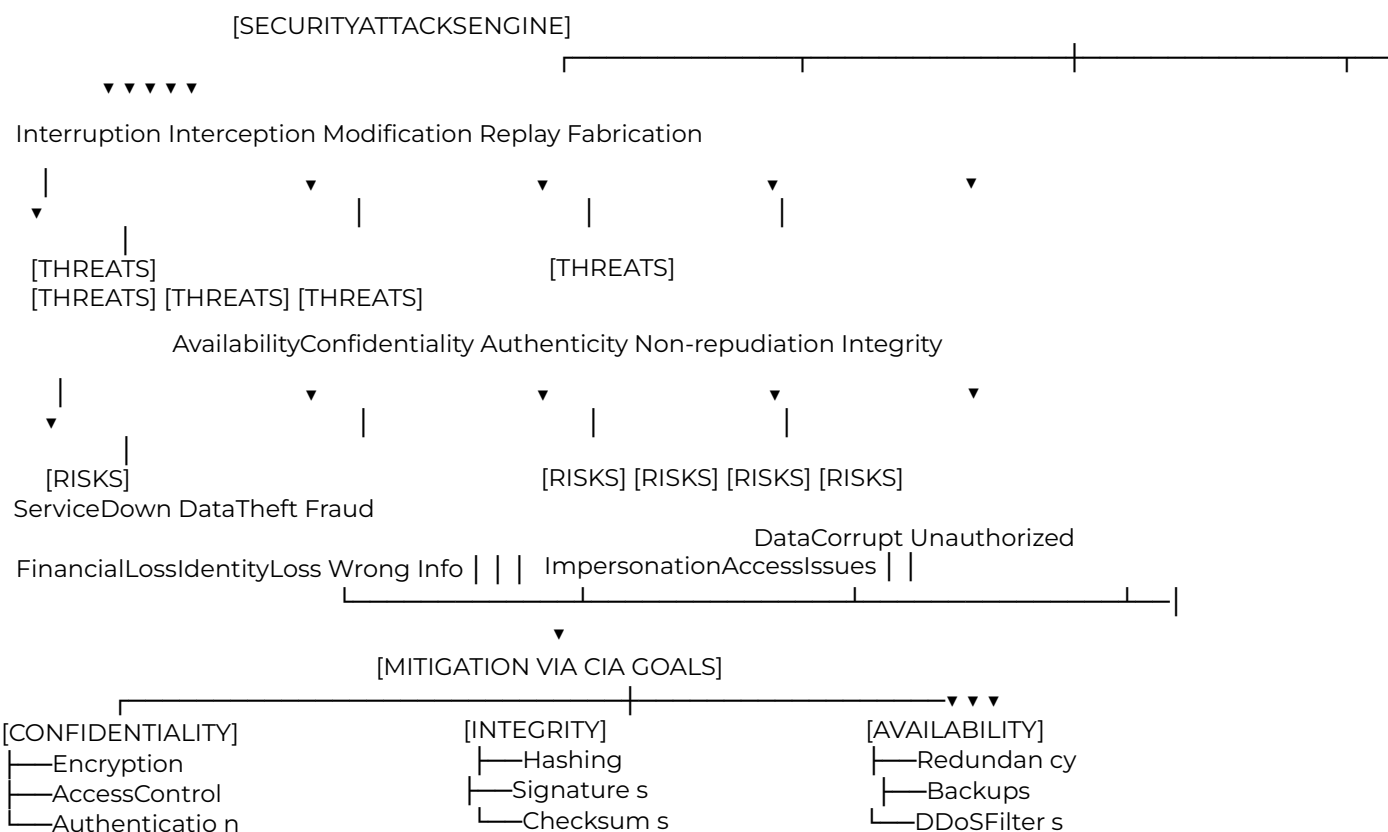
RUBRICS FOR CALCULATION:

QUESTION 1: SECURITY ATTACKS, THREATS, RISKS, AND SECURITY

GOALS 1. Introduction to Security Architecture

In network security engineering, understanding the split between security attacks, operational threats, real world risks, and system security goals is vital for building resilient systems. An attack represents an active or passive malicious action taken against a resource. A threat is the vulnerability or potential danger exposed by that action, while a risk defines the concrete operational or financial loss suffered by the company. Security goals (Confidentiality, Integrity, and Availability) provide the structural defense rules needed to counter these attacks.

2. Concept Map



3. Detailed Explanation of Attack-to-Risk Causal Chains

Every security failure moves through a clear chain of events where a technical action turns into business damage:

- Interruption:** This attack targets system connectivity. By breaking communication lines or overloading servers, the adversary triggers an ****Availability Threat****. The business risk includes full service downtime, lost customer interactions, and immediate drop in digital storefront revenue.
- Interception:** This is a passive monitoring action where an unauthorized party captures private data. It creates a ****Confidentiality Leak Threat****. The associated risks include intellectual property theft, customer data exposure, and severe regulatory fines under laws like GDPR.
- Modification:** This active attack alters intercepted data packets before they reach their destination. This acts as an ****Integrity Compromise Threat****. The business risks include incorrect accounting ledgers, corrupted databases, and wrong operational processing updates.
- Fabrication:** The attacker inserts fake objects or messages into a communication stream, representing an ****Authenticity Breach Threat****. The business risks include phishing financial scams, fraudulent billing requests, and complete credential theft.
- Replay Attack:** The adversary captures a valid data transmission packet and re-transmits it later to duplicate an action. This represents a ****Non-repudiation and Validation Threat****, causing risks like duplicate banking withdrawals or unauthorized system entrance.

4. Mapping Matrix Table

Attack Type	[MITIGATION VIA CIA GOALS]			
	Confidentiality	Integrity	Availability	Context
Interruption				Loss of Availability
Interception				Loss of Confidentiality
Modification				
Replay				
Fabrication				

Loss of Integrity	financial loss	Duplicate sessions, transaction fraud	Integrity via SHA-256 hashes & digital signatures
	Data theft, privacy litigation		
Authenticity		CIA Mitigation Target	Authentication via Multi-Factor Tokens
Loss of Non repudiation	Database corruption, wrong calculations	Availability via load-balancers & backups	
	Identity scams, illegal fund transfer	Confidentiality via AES/RSA encryption	Validation via unique nonces & timestamps
Business Risk			
Downtime, missed SLAs,			

5. Real-World Example

A notable real-world example occurred during a major ransomware strike on a global logistics firm. The attackers launched an **Interruption** attack by encrypting the core database servers, making them completely unusable. This created a complete **Loss of Availability Threat**, as tracking systems went dark overnight. The company suffered huge **Financial and Operational Risks**, spending millions on recovery and losing customer trust. The firm mitigated future threats by building separate off-site storage backups and automated failover groups to secure system **Availability**.

6. Security Analysis and Discussion

Mitigating modern risks requires transitioning from reactive patching to a proactive defense-in-depth framework using the CIA Triad. To ensure ****Confidentiality****, organizations must encrypt all sensitive traffic at rest and in transit using secure protocols like TLS.****Integrity**** defenses must rely on cryptographic hash functions like SHA-256 alongside digital signatures. If an attacker uses a **Modification** attack to alter a single bit of data in transit, the calculated hash at the receiving end will mismatch, causing the system to drop the packet instantly.****Availability**** requires removing single points of failure by using distributed cloud hosting, load balancers, and redundant hardware arrays. This ensures that even if an **Interruption** attack knocks offline a primary node, traffic is re-routed immediately to preserve service runtime.

7. Conclusion

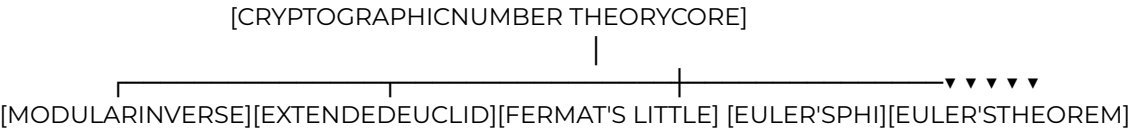
Security attacks directly undermine system health by converting vulnerabilities into operational threats and corporate risks. By building a balanced security infrastructure focused on Confidentiality, Integrity, and Availability, network engineers can intercept these attack pathways and maintain secure system operations.

QUESTION 5: NUMBER THEORY CONCEPTS IN CRYPTOGRAPHY

1.Introduction to Cryptographic Number Theory

Modern public-key cryptography architectures rely heavily on abstract number theory. Asymmetric algorithms like RSA, Diffie-Hellman, and the Digital Signature Algorithm (DSA) do not depend on simple text shifting or scrambling blocks. Instead, they protect data using complex, one-way mathematical functions that are easy to compute in one direction but incredibly difficult to reverse unless you hold a specific private key ledger.

2. Concept Map



key matrix) inverses) tests) spaces) asymmetric ciphers)

3. Comprehensive RSA Number Theory Matrix		
	Active Number Theory Component	Primary Cryptographic
Asymmetric Step Context		
Prime Selection	Fermat's Little Theorem Matrix	

Function Verifies big random inputs are prime numbers

Modulus Generation	$\phi(n) = (p-1)(q-1)$	loop limit for key generation	inverse value
Euler's Phi Function	Calculates the co-prime	Calculates the modular	
Private Key Derivation	Extended Euclidean Algorithm $\text{mod } \phi(n)$	$d = e^{-1}$	
	Euler's Theorem Generalization	, restoring	
	Guarantees that plaintext		
Text Decryption	$M^{ed} \text{ mod } n \equiv M$		

6. (c) Critical Concept Justification

Among all the number theory concepts discussed,**Euler's Theorem is the most critical cornerstone for public-key cryptography**. While tools like the Extended Euclidean Algorithm are useful for speed optimizations, Euler's Theorem provides the core mathematical justification that makes asymmetric key pairs possible.

The central challenge of public-key cryptography is creating a secure "trapdoor function": a mathematical operation that allows anyone to easily encrypt data using a public key, but ensures only the authorized holder of the private key can reverse the operation. Euler's Theorem solves this problem through the equation:

$$a^{\phi(n)} \equiv 1 \pmod n$$

This relationship allows us to construct a key pair where the public component *e* and private component *d* form an identity loop: *e imes d = 1 + k |cdot phi(n)*. When we raise data to this combined power, the message transforms as follows:

$$M^{ed} \equiv M^{1 + k \cdot \phi(n)} \equiv M \cdot (M^{\phi(n)})^k \equiv M \cdot (1)^k \equiv M \pmod n$$

This elegant mathematical behavior is what allows a receiver to decrypt intercepted ciphertext using a private key. Without the mathematical foundation provided by Euler's Theorem, systems would have no way to establish secure public-key encryption channels, leaving global networks without modern asymmetric defense options.

7. Conclusion

Number theory provides the essential mathematical frameworks needed to secure modern asymmetric ciphers. By linking prime numbers, modular inverses, and Euler's Theorem, network systems can generate secure public-private key pairs that protect global enterprise data transmissions from sophisticated interception threats.

Criteria Weight	Level 4 – Exemplary	Level 3 – Proficient	Developing	Beginning
Understanding of Concepts	25 Demonstrates thorough understanding and effectively integrates multiple integration	Good understanding with proper partial integration concepts	Basic understanding; limited or Level 1 – integrate	Poor understanding; unable to

Experiment concepts/technology	gies Well-planned, systematic of most concepts	implementat Basic implementati	Incorrect or incomplete al Design on	30 implementation ion with on with errors implementati
	Correct	with optimal solution minor issues or		
Use of Tools	use of tools with correct execution			
Analysis of Results	Good analysis with reasonable interpretation Organized documentati on with necessary details			
Documentat ion	Limited or inconsistent use of tools			
20 15 10	Basic analysis with limited insights			
Effective and advanced use of tools/technologi es with proper configuration	Basic documentatio n with missing details			
Insightful analysis with accurate interpretation and validation of results	Incorrect or minimal use of tools Poor or incorrect analysis of results			
Clear, well structured documentation with diagrams, code, and results	Poor or incomplete documentatio n			
Appropriate				